

Alexander Ching

PHD GRADUATE STUDENT RESEARCHER

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Education

University of Washington

Seattle, WA

PHD (POST-QUALS) IN COMPUTER SCIENCE AND ENGINEERING

Expected 2027/28

- My research is a mix of sensing, applied machine learning, hardware design, and signal processing.
- I focus on biological and physiological signals for applications in health and interaction.
- Advised by Dr. Shwetak Patel, Ubiquitous Computing Lab

University of Washington

Seattle, WA

MS IN COMPUTER SCIENCE AND ENGINEERING

2023

- Pre-Hypertension Detection via Finger Photoplethysmography
- Advised by Dr. Shwetak Patel and Dr. Luis Ceze

University of Washington

Seattle, WA

BS IN ELECTRICAL ENGINEERING, BS IN MECHANICAL ENGINEERING

2014

- NASA Microgravity University 2012, 2014
- Human Powered Submarine Team

Professional Experience

Samsung Research America

Plano, TX

RESEARCH INTERN - NEXT-GEN CONNECTIVITY & SENSING RESEARCH

Jun 2025 - Sep 2025

- Research on low power eye tracking and facial expression recognition solutions, winning first place at intern poster session.
- Designed and built a XR prototype; I implemented all parts of the stack: hardware, firmware, and model development.
- Developed user data collection procedure, ML models (XGBoost, CNN, MLP), and real time inference code.

Meta Reality Labs Research

Redmond, WA

SENIOR ELECTRICAL ENGINEER - DISPLAYS SYSTEMS RESEARCH GROUP

Jan 2022 - Oct 2023

- Electronics design for prototype VR headsets, perception systems, and mixed reality pass-through applications.
- Worked with MIPI CPHY displays, DPHY cameras, FPGA based camera aggregation, systems architecture design.
- Lead EE on Flamera, SIGGRAPH 2023 Emerging Technologies Best in Show Award Winner. US Patent 20240251176A1

Microsoft

Redmond, WA

ELECTRICAL ENGINEER II - MIXED REALITY

Dec. 2019 - Jan 2022

- EE lead for Augmented/Virtual reality controllers to be used with future Mixed Reality devices.
- AR/VR hardware concept prototyping for both production and in collaborations with research teams.
- Worked with overseas contract vendors for larger design programs.

Magic Leap

Seattle, WA

LEAD RESEARCH ENGINEER - ADVANCED TECHNOLOGIES

Oct. 2016 - May 2019

- Led electrical and systems architecture design for next generation augmented reality prototypes.
- Worked directly on full cycle EE develop from schematics to final systems integration of various camera and display systems.
- Managed internal collaborations external engineering contractors, and third-party vendors/suppliers.

Microsoft Research

Redmond, WA

ELECTRICAL ENGINEER - HARDWARE LAB

Jan. 2015 - Oct. 2016

- Lead hardware engineer for Project Premonition, a robotic insect trap for tracking infectious diseases. US Patent 10966420.
- Electrical and Mechanical design and prototyping in partnership with MSR researchers working on VR/AR, Robotics, HCI, Cyber-physical Systems, Audio, Healthcare, Sensing, Embedded Systems, etc.
- Developed programs from concept to prototype ranging from single day rush to multi-year projects.

Valve

Bellevue, WA

HARDWARE ENGINEER CONTRACTOR - VR TEAM

Mar. 2014 - Jan. 2015

- Hardware prototyping of electrical and mechanical systems for development of VR tracking systems.
- Designed electro-mechanical device for longevity testing of the Steam controller.
- Helped build the original batch of lighthouse base stations.

NASA Ames Research Center, Intellectual Ventures Lab, Cisco Systems, Ball Aerospace

Various

ENGINEERING UNDERGRADUATE INTERNSHIPS/CO-OPS

Research Projects

University of Washington - CSE

Seattle, WA

ADVISOR: DR. SHWETAK PATEL

- **Blood Pressure Ring** (Ongoing): We are developing a ring form factor device that measures all day blood pressure using the oscillometric method and photoplethysmography (ppg) to track relative changes of an individual's blood pressure continuously. Currently designing a compact ring based device for a longitudinal data collection study.
- **Tuberculosis Classification** (Ongoing): We are working with clinical partners at UW Medicine, and Kenya Medical Research Institute on a pipeline to classify TB, and TB infectiousness based on coughs obtained both from the clinic and ambulatory wearable devices. We are creating an audio pipeline to clean the data, extract coughs, and train/test a model for classification. I setup the data collection study on the wearable devices that is currently in progress with our partners in Kenya. I also wrote the majority of the grant proposal for the UW Population Health Department's Application of AI to Grand Challenges in Population Health Initiative which awarded \$75k towards this research.
- **Patient Extubation Failure Classification** (Ongoing): Patients with severe acute brain injury who are extubated will often need to be re-intubated. This project explores the links between respiratory signals and extubation failure. I am working on a digital stethoscope for data collection on patients, the signal processing to do feature extraction, and development of ML models to link biomarkers with extubation prognosis. I work with our clinical partners at UW Medicine and Harborview Medical Center in order to implement our user study. I contributed to the grant proposal for the UW Institute for Medical Data Science Pilot Award for which \$25k was awarded for this project.
- **Ring and Wrist Based Hand Tracking** (Completed - In Review): Exploration of mechanically mediated sensing using strain sensors embedded into novel structures in a ring and wrist worn device to track user gestures such as pinches, rotations, etc. Our design grants high dynamic range with low noise and the ability to target and optimize sensing to specific motions.
- **Ventriculoperitoneal Shunt failure detection** (Ended): VP shunts are used for hydrocephalus treatment to help drain fluid from the brain cavity, but have a failure rate over 90% at some point in their usage due to blockages and other failures. We looked at ways to non-invasively detect these failures with our clinical partners at UW Medicine. We built an NFC based non-battery pressure sensing circuit with an Android app to interface with it.
- **Hypertension Detection Using PPG Oscillometric Method** (Completed - Qualification Exam Project): Cardiovascular disease is the leading cause of death in the US. I built a finger based ppg sensor coupled with a force sensor in order to calculate one's blood pressure using the oscillometric method. The data was collected from participants and analyzed using Matlab.

Publications

Grace Kuo, Eric Penner, Seth Moczylowski, **Alexander Ching**, Douglas Lanman, and Nathan Matsuda. 2023. Perspective-Correct VR Passthrough Without Reprojection. In ACM SIGGRAPH 2023 Conference Proceedings (SIGGRAPH '23). Association for Computing Machinery, New York, NY, USA, Article 15, 1–9. <https://doi.org/10.1145/3588432.3591534>.

Alexander Ching. Designing Embedded Hardware A Guide for Students & New Grads. 2025. https://atching.github.io/files/Intro_to_Embedded_Hardware.pdf

IN REVIEW

Alexander Ching, Anum Ali, Abhishek Sehgal, Tao Chen, Viet Nguyen, Rebal Jurdi, Neha Dawar, Yuming Zhu, Boon Loong Ng. 2026. XR Prototyping with a Modular Hardware Sensing Platform. Submitted to ISWC May 2026.

Alexander Ching, Xizhe Hao, Yi-Hsuan Ho, Ping Wang, Shwetak Patel. 2026. Beyond Wear and Tear: Mechanically Mediated Sensing for Wearable Input. Submitted to IMWUT May 2026.

Teaching Experience

AUT 2026	CSE 475 - Embedded Systems Capstone , Graduate Teaching Assistant
AUT 2025	CSE 475 - Embedded Systems Capstone , Graduate Teaching Assistant
SPR 2025	CSE 475 - Embedded Systems Capstone , Graduate Teaching Assistant
AUT 2024	CSE 475 - Embedded Systems Capstone , Graduate Teaching Assistant
SPR 2024	CSE 474 - Introduction to Embedded Systems , Graduate Teaching Assistant
WIN 2024	CSE 461 - Computer Networking , Graduate Teaching Assistant
AUT 2023	CSE 475 - Embedded Systems Capstone , Graduate Teaching Assistant
AUT 2022	CSE 475 - Embedded Systems Capstone , Graduate Teaching Assistant

Patents

Microsoft Research

Redmond, WA

ELECTRICAL ENGINEER

- Insect Trap, US 10966420 - Issued Apr 6, 2021 US

Meta Reality Labs Research

Redmond, WA

ELECTRICAL ENGINEER

- Perspective-Correct Passthrough Architectures for Head-Mounted Displays, US 20240251176A1 - Issued Jul 25, 2024 US

Mentoring

9/25 - 6/26	Arya Cholkar , CSE Undergraduate Student, University of Washington	
9/25 - 6/26	Jyothi Panchapagesan , Undergraduate Student, University of Washington	
4/26 - 6/26	Andi Cai , ECE Masters Student, University of Washington	
9/25 - 6/26	Yuzhi Fan , CSE Undergraduate Student, University of Washington	
9/25 - 6/26	George Evans Daenuwy , ECE Undergraduate Student, University of Washington	
11/24 - 6/26	Zach (Xizhe) Hao , ECE Masters Student, University of Washington	Notre Dame PhD
2/25 - 6/26	Mingzhuo Ma , ECE Masters Student, University of Washington	UMich PhD
9/25 - 5/26	Cidney Ho , Technology Innovation Masters Student, University of Washington	
9/25 - 3/26	June (Junhyeok) Lee , Bio-Engineering Masters Student, University of Washington	Google
9/24 - 6/25	Peter (Ping) Wang , CSE Undergraduate Student, University of Washington	
9/24 - 6/25	Parum Misri , CSE Undergraduate Student, University of Washington	Amazon

Skills

Software	Python, PyTorch, C, Visual Studio Code, AWS
Analysis	Signal Processing, Matlab, Applied Machine Learning, Applied Deep Learning
Electrical	Altium, Cadence, LTSpice, PCB Design, Soldering
Mechanical	SolidWorks, 3D Printing, CNC Manufacturing
Misc	LaTeX, Photography